



Information Gathering

BIS601

Information Gathering

- **Gather information** on what the system should do from as many sources as possible
 - Users of the current system
 - Observing users
 - Reports, forms, and procedures

Information

- Information collected from conversations with or observations of users: interview transcripts, notes from observations, meeting minutes
- Existing written information: business mission and strategy statements, sample business forms and reports and computer displays, procedure manuals, job descriptions, training manuals, flowcharts and documentation of existing systems, consultant reports

Information

- Computer-based information: reports of existing systems, displays and reports from system prototypes

Understanding

- The business objectives that drive what and how work is done
- The information people need to do their jobs
- The data (definition, volume, size, etc.) handled within the organization to support the jobs
- When, how, and by whom or what data are moved, transformed, and stored

Understanding

- The sequence and other dependencies among different data-handling activities
- The rules governing how data are handled and processed
- Policies and guidelines that describe the nature of the business and the market and environment in which it operates
- Key events affecting data values and when these events occur

Analysis Paralysis

- Huge amount of information, especially if the scope of the system under development is broad
- Time, human effort: quite expensive
- Too much analysis is not productive
- Analysis paralysis: a systems development project that has become bogged down in an abundance of analysis work

Analysis Paralysis

- Focus more on the system to be developed than on the current system
- Prototyping, continual user involvement, usage-centered design

Interview

- One of the primary ways
- Interviewing people about their work, the information they use to do it, the types of information processing that might supplement their work
- Understand organizational direction, policies, expectations managers have on the units they supervise, other nonroutine aspects of organizational operations

Interview

- During interviewing you will gather facts, opinions, and speculation and observe body language, emotions, and other signs of what people want and how they assess current systems

Interview: What to do

- Plan the interview
 - Prepare interviewee: appointment, priming questions
 - Prepare checklist, agenda, and questions
- Listen carefully and take notes (record if permitted)
- Review notes within 48 hours of interview
- Be neutral
- Seek diverse views

Plan the Interview

- Set up the appointment
- Explain general nature of the interview
- Think about what you need to find out and write down your questions
- Prepare an interview guide or checklist

Interview Outline	
Interviewee: <i>Name of person being interviewed</i>	Interviewer: <i>Name of person leading interview</i>
Location/Medium: <i>Office, conference room, or phone number</i>	Appointment Date: Start Time: End Time:
Objectives: <i>What data to collect On what to gain agreement What areas to explore</i>	Reminders: <i>Background/experience of interviewee Known opinions of interviewee</i>
Agenda: Introduction Background on Project Overview of Interview Topics to Be Covered Permission to Record Topic 1 Questions Topic 2 Questions ... Summary of Major Points Questions from Interviewee Closing	Approximate Time: 1 minute 2 minutes 1 minute 5 minutes 7 minutes ... 2 minutes 5 minutes 1 minute
General Observations: <i>Interviewee seemed busy probably need to call in a few days for follow-up questions because he gave only short answers. PC was turned off—probably not a regular PC user.</i>	
Unresolved Issues, Topics Not Covered: <i>He needs to look up sales figures from 1999. He raised the issue of how to handle returned goods, but we did not have time to discuss.</i>	
Interviewee:	Date:

Interviewee:	Date:
Questions:	Notes:
<i>When to ask question, if conditional</i> Question: 1 Have you used the current sales tracking system? If so, how often?	Answer Yes, I ask for a report on my product line weekly. Observations Seemed anxious—may be overestimating usage frequency. <i>If yes, go to Question 2</i>
Question: 2 What do you like least about the system?	Answer Sales are shown in units, not dollars. Observations System can show sales in dollars, but user does not know this.

Choosing Interview Questions

- Decide what mix and sequence of open-ended and closed-ended questions you will use
- **Open-ended questions:** questions that have no prespecified answers
 - For information for which you cannot anticipate all possible responses
 - Interviewee encouraged to talk about whatever interests him within the general bound of the question

Choosing Interview Questions

- You must react quickly to the answers and determine whether or not any follow-up questions are needed for clarification or elaboration
- Previously unknown information can surface
- Time-consuming, difficult to summarize

Choosing Interview Questions

- **Closed-ended questions:** questions that ask those responding to choose from among a set of specified responses
 - True or false, multiple choice, rating, ranking
 - The major answers to questions are well known
 - You can include an “other” option to encourage the interviewee to add unanticipated responses
 - Useful information may be overlooked as the respondent tries to make a choice instead of providing his best answer

Guidelines

- Do not phrase a question in a way that implies a right or wrong answer
 - E.g., “Should the system continue to provide the ability to override the default value, even though most users now do not like the feature?”
- Listen very carefully to what is being said; take careful notes

Guidelines

- Once the interview is over, type up your notes within 48 hours
 - Additional questions, unclear points that need clarification
 - You may send a copy of your notes to the interviewee to check your notes for accuracy
 - Thank the interviewee for his time

Guidelines

- Do not set expectations about the new system unless you are sure these features will be parts of the delivered system
- Seek a variety of perspectives from the interview

Interviewing Groups

- A series of interviews may turn up inconsistent information
- Several follow-up, additional interviews
- Group interview: several key people are interviewed as once
- Much more effective use of time
- Allow people to hear the opinions of other key people and give them the opportunity to agree or disagree with their peers

Interviewing Groups

- The comment of one person might cause another person to say, “That reminds me of ...”
- Difficult to schedule

Nominal Group Technique

- A facilitated process that supports idea generation by groups
 - At the beginning of the process, group members work alone to generate ideas
 - The ideas are then pooled and discussed
- Apply to problems with the existing system or ideas of new features in the system being developed

Nominal Group Technique

- The end result would be a list of either problems or features that group members had generated and prioritised
- High level of ownership

Observation

- Because people cannot always be trusted to reliably interpret and report their own actions, you can supplement what people tell you by watching what they do or obtaining relatively objective measures of how people behave in work situations

For example, one possible view of how a hypothetical manager does her job is that a manager carefully plans her activities, works for long periods of time and consistently on solving problems, and controls the pace of her work. A manager might tell you that is how she spends her day. When **Mintzberg (1973)**  observed how managers work, however, he found that a manager's day is actually punctuated by many, many interruptions. Managers work in a fragmented manner, focusing on a problem or on a communication for only a short time before they are interrupted by phone calls or e-mail or visits from their subordinates and other managers. An information system designed to fit the work environment described by our hypothetical manager would not effectively support the actual work environment in which that manager finds herself.

As another example, consider the difference between what another employee might tell you about how much he uses e-mail and how much e-mail use you might discover through more objective means. An employee might tell you he is swamped with e-mail messages and that he spends a significant proportion of his time responding to e-mail. However, if you were able to check electronic mail records, you might find that this employee receives only three e-mail messages per day on average, and that the most messages he has ever received during one eight-hour period is 10. In this case, you were able to obtain an accurate behavioral measure of how much e-mail this employee copes with without having to watch him read his e-mail.

Lost Soft Drink Sales

A systems analyst was quite surprised to read that sales of all soft-drink products were lower, instead of higher, after a new delivery truck routing system was installed. The software was designed to reduce stock-outs at customer sites by allowing drivers to visit each customer more often using more efficient delivery routes.

Confused by the results, management asked the analyst to delay a scheduled vacation, but he insisted that he could look afresh at the system only after a few overdue days of rest and relaxation.

Instead of taking a vacation, however, the analyst called a delivery dispatcher he interviewed during the design of the system and asked to be given a route for a few days. The analyst drove a route (for a regular driver who was actually on vacation), following the schedule developed from the new system. What the analyst discovered was that the route was very efficient, as expected; so, at first, the analyst could not see any reason for lost sales.

During the third and last day of his “vacation,” the analyst stayed overtime at one store to ask the manager if she had any ideas why sales might have dropped off in recent weeks. The manager had no explanation but did make a seemingly unrelated observation that the regular route driver appeared to have less time to spend in the store. He did not seem to take as much interest in where the products were displayed and did not ask for promotional signs to be displayed, as he had often done in the past.

From this conversation, the analyst concluded that the new delivery truck routing system was, in one sense, too good. It placed the driver on such a tight schedule that he had no time left for the “schmoozing” required to get special treatment, which gave the company’s products an edge over the competition.

Without firsthand observation of the system in action gained by participating as a system user, the analyst might never have discovered the true problem with the system design. Once time was allotted for not only stocking new products but also for necessary marketing work, product sales returned to and exceeded levels achieved before the new system had been introduced.

Observation

- Observation can cause people to change their normal operating behavior
- Cannot be continuous; you receive only a snapshot image of the person or task you observe, which may not include important events or activities
- Exactly which people or sites to observe at what time is a difficult selection problem

Procedures and Document Analysis

- Examining system and organizational documentation to discover more details about current systems and the organization these systems support

Procedures and Document Analysis

- A written work procedure for an individual or a work group
 - Describe how a particular job or task is performed, including data and information that are used and created in the process of performing the job

Procedures and Document Analysis

- The difference between formal systems and informal systems
 - Formal system: the official way a system works as described in organizational documentation
 - Informal system: the way a system actually works
- Understand both formal and informal systems; each provides insight into information requirements

Procedures and Document Analysis

- Business form
 - Explicitly indicate what data flow in and out of a system and which are necessary for the system to function
 - Allow you to determine the characteristics of the data that are actually used by the application

Procedures and Document Analysis

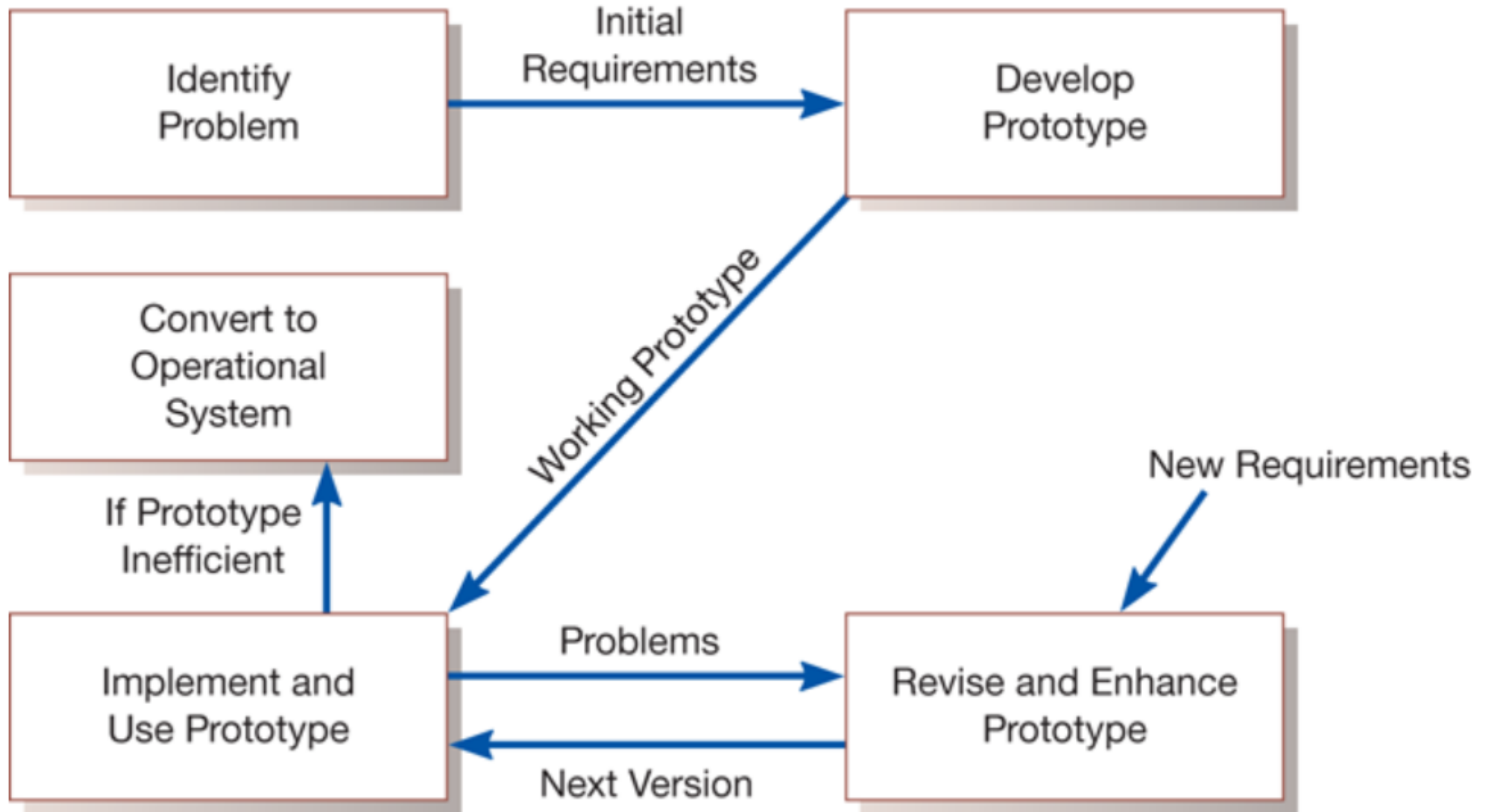
- Report generated by current systems
 - As the primary output for some types of systems, a report enables you to work backward from the information on the report to the data that must have been necessary to generate them

Procedures and Document Analysis

- Documentation of current information systems
 - How they were designed and how they work
 - Everything from the flowcharts to data dictionaries to user manuals

Prototyping

- An iterative process of systems development in which requirements are converted to a working system that is continually revised through close collaboration between an analyst and users



Prototyping

- The prototype captures only parts of the system requirements: typical cases, not exception handling
- Also some design: database access, data integrity, system security, scalability, multiuser support, multiplatform support

Prototyping

- Drawbacks

- Prototypes have a tendency to avoid creating formal documentation of system requirements, which can then make the system more difficult to develop into a fully working system.
- Prototypes can become very idiosyncratic to the initial user and difficult to diffuse or adapt to other potential users.
- Prototypes are often built as stand-alone systems, thus ignoring issues of sharing data and interactions with other existing systems, as well as issues with scaling up applications.
- Checks in the SDLC are bypassed so that some more subtle, but still important, system requirements might be forgotten (e.g., security, some data entry controls, or standardization of data across systems).

Characteristic	Observation	Document Analysis
Information Richness	High (many channels)	Low (passive) and old
Time Required	Can be extensive	Low to moderate
Expense	Can be high	Low to moderate
Chance for Follow-Up and Probing	Good: probing and clarification questions can be asked during or after observation	Limited: probing possible only if original author is available
Confidentiality	Observee is known to interviewer; observee may change behavior when observed	Depends on nature of document; does not change simply by being read
Involvement of Subject	Interviewees may or may not be involved and committed depending on whether they know if they are being observed	None, no clear commitment
Potential Audience	Limited numbers and limited time (snapshot) of each	Potentially biased by which documents were kept or because document was not created for this purpose